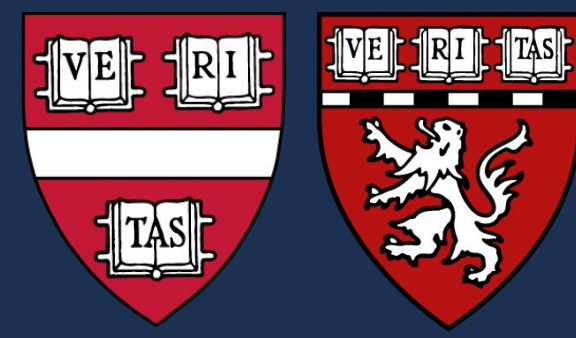


ARC: An algorithmic approach to remove race correction in clinical equations

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MOTIVATION

- Race Stratification in Clinical Equations:** Increasing scrutiny on race-stratified clinical algorithms for kidney function, lung function equations, and cardiovascular disease among other areas has led to interest in race-free alternatives^{1,2}
- Pulmonary Function Tests (PFTs):** PFTs assist in diagnosis and evaluation of lung diseases by comparing observed spirometry values to expected reference values, historically stratified by race^{3,4}
- Implications of Race Stratification:** Race stratification in PFTs projects an expected 20% + in lung function among Black individuals and a 4-6% + in Asians⁴, leading to significant disparities in medical diagnoses, lung transplant priority, preoperative evaluation, insurance premiums, and disability compensation^{5,6}
- Limitations with ATS Guideline:** In 2022, the American Thoracic Society (ATS) recommended GLI-Global, race-neutral equations trained with equally weighted groups; however, these equations do not capture the underlying cause of lung-function variation among populations^{7,8}

Reference PFT ~ f(Age, Sex, Height, *Race)

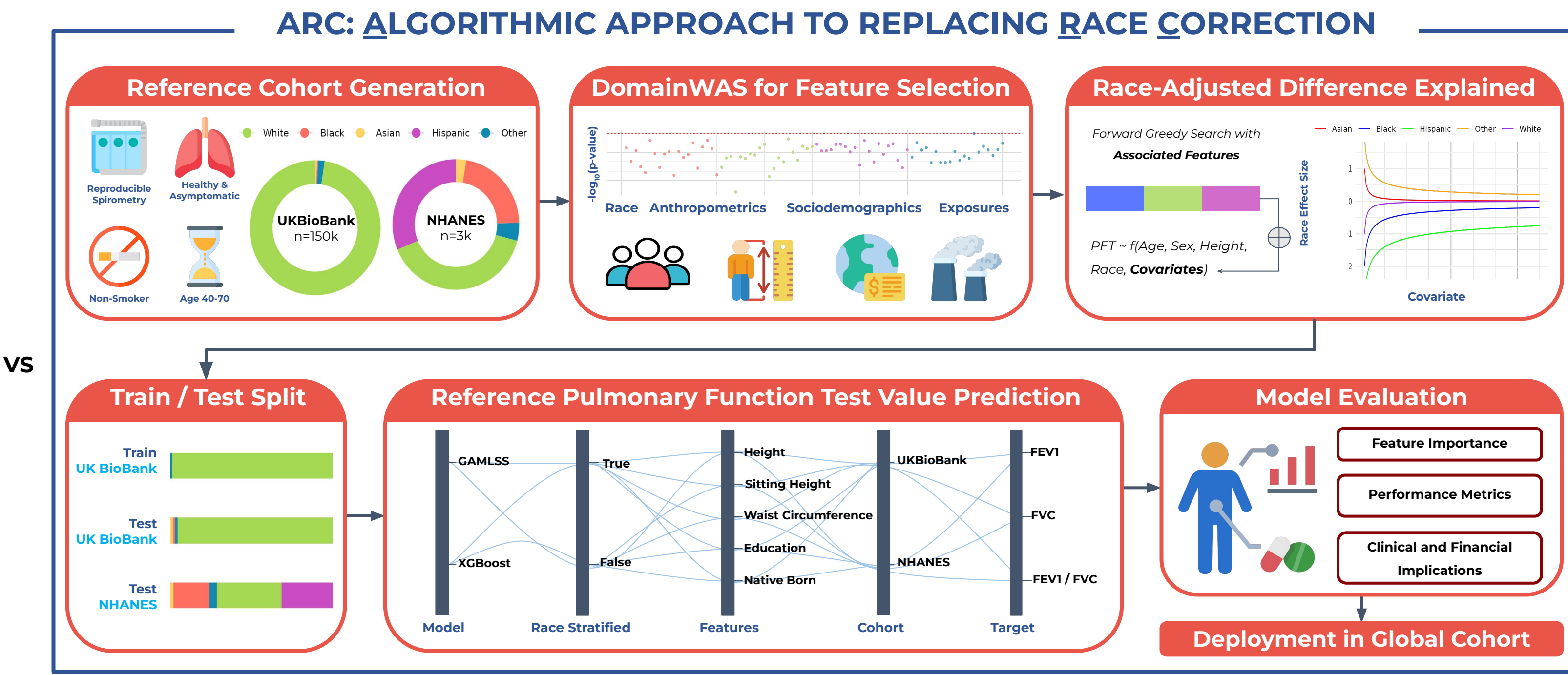
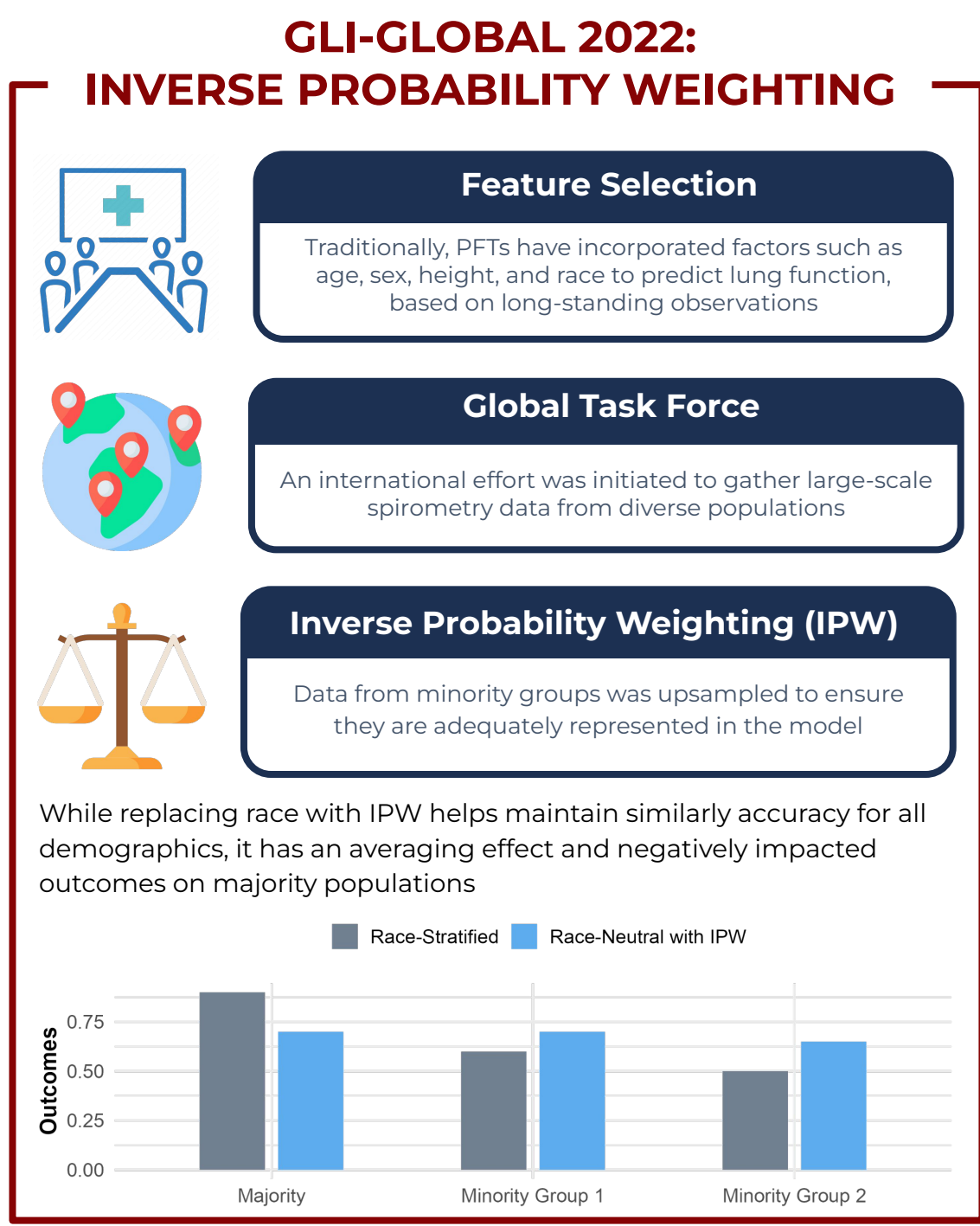
	White	Black	Asian
Age	30	30	30
Sex	Male	Male	Male
Height (cm)	175	175	175
Observed FVC (L)	4	4	4
Interpretation	Low	Normal	Borderline

Limitations with Race Correction

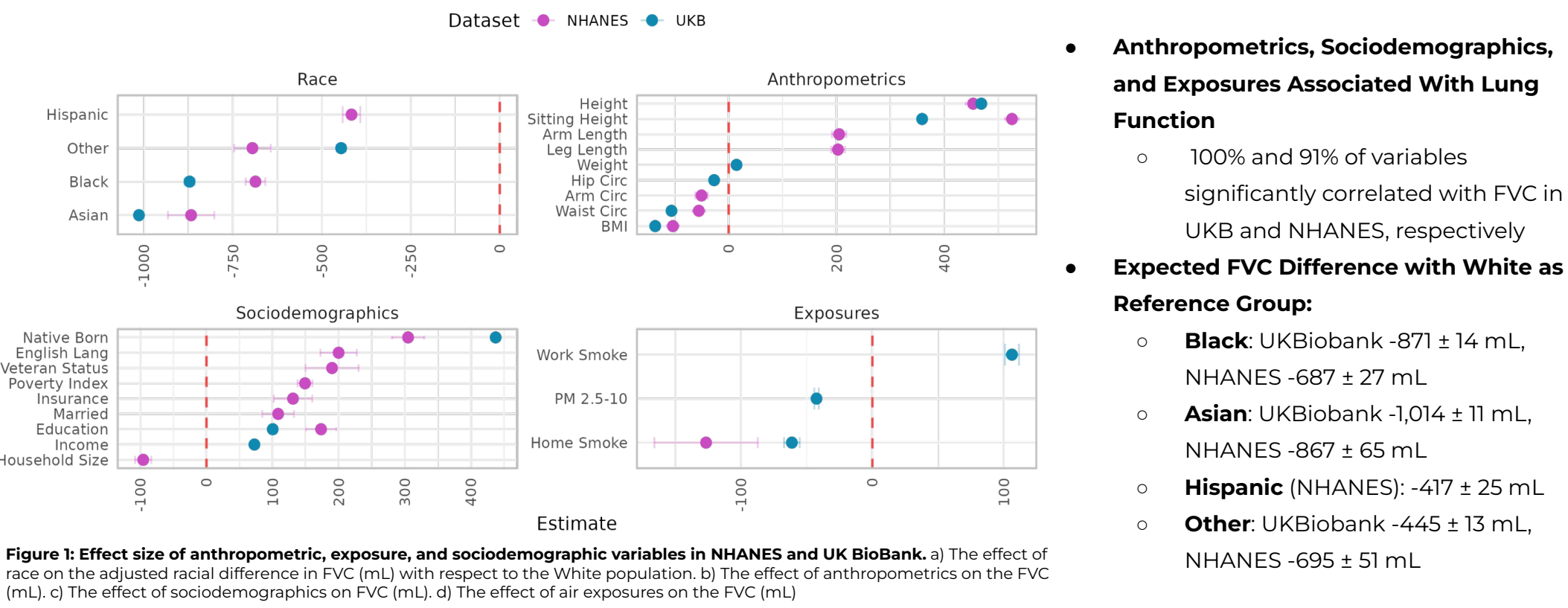


Approach: We developed ARC, an algorithmic approach to replace race correction in clinical equations, which systematically identifies factors that may explain race-related variation and utilizes such features to promote fairness, accuracy, and robustness.

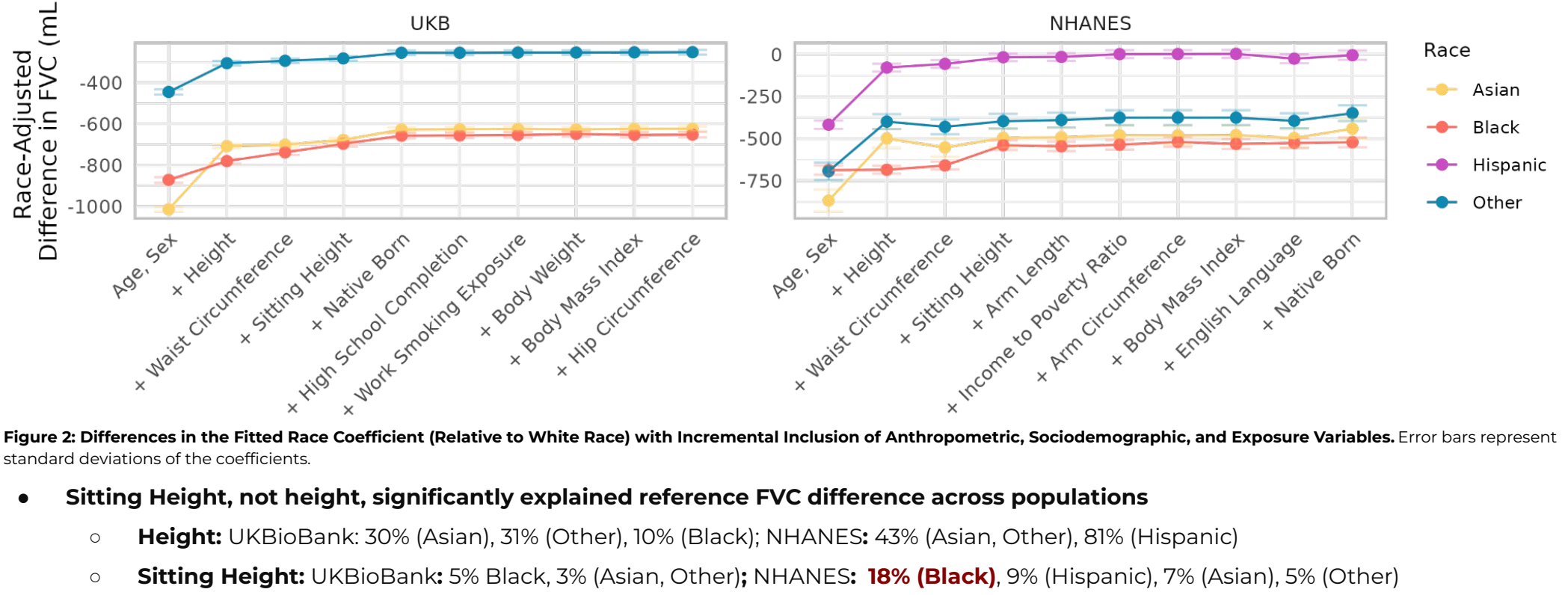
WORKFLOWS FOR REMOVING RACE IN PULMONARY FUNCTION TESTS



ASSOCIATIONS WITH REFERENCE LUNG FUNCTION



RACE-ADJUSTED DIFFERENCE EXPLAINED



COMPARISON WITH RECOMMENDED EQUATIONS

- ARC outperformed GLI-Global in reference FVC prediction across cohorts**
 - Additional features improve performance for all groups, despite no inverse probability weights
- Anthropometrics, not sociodemographics, improved FVC prediction across cohorts**
 - Sitting height:** 3% + MAE 3% (UK Biobank) and 12% (NHANES)
 - Waist circumference:** 16% + Black MAE (NHANES)
 - Native-born:** 9% + MAE (UKB); + MAE (NHANES)
- Performance gap + between Non-White and White subgroups**
 - GLI-Global:** 44% (UKBiobank) and 35% (NHANES)
 - + Sitting height:** 39% (UK Biobank) and 26% (NHANES)
 - + Waist circumference:** 39% (UK Biobank) and 8% (NHANES)
- Race-neutral model outperformed race-based models in external validation, showcasing robustness to domain shift**

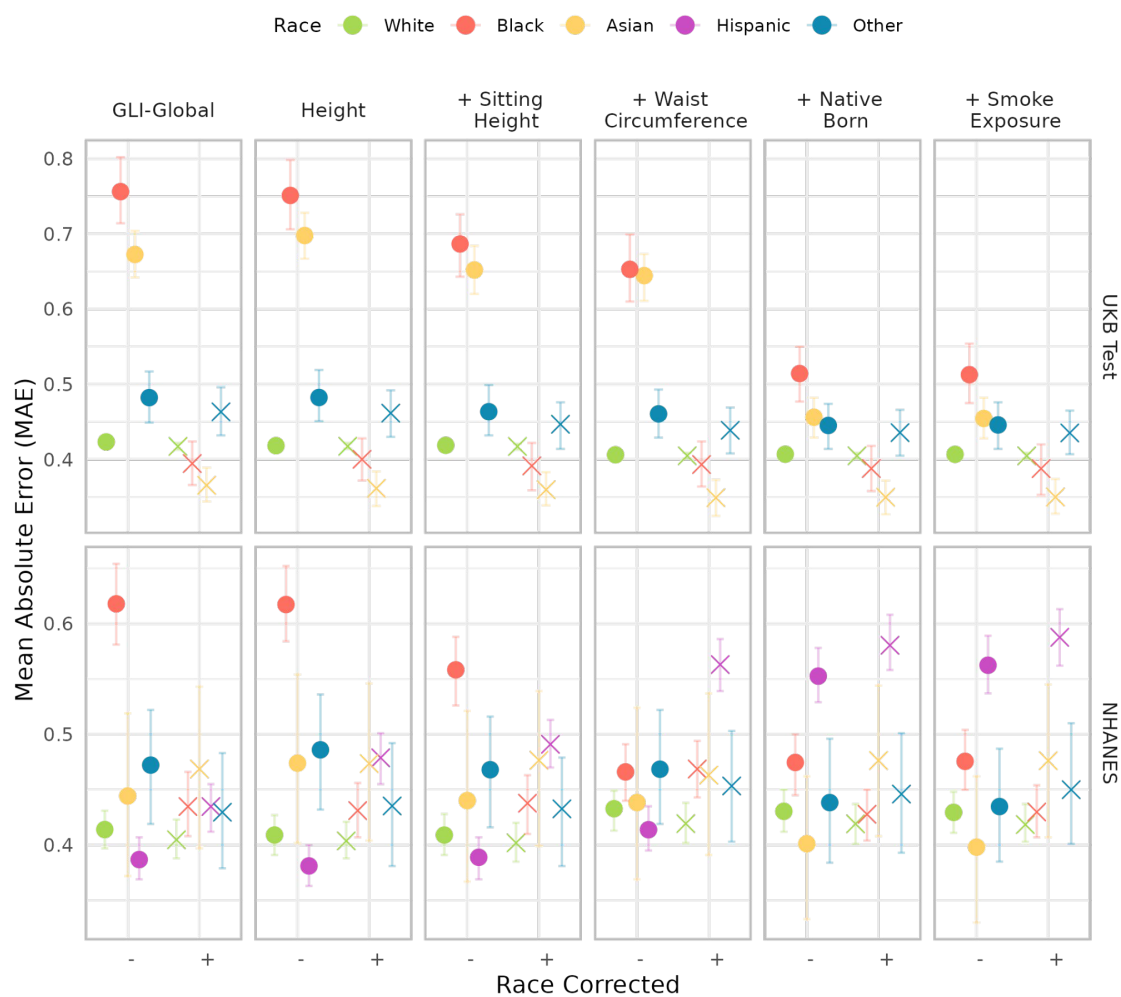
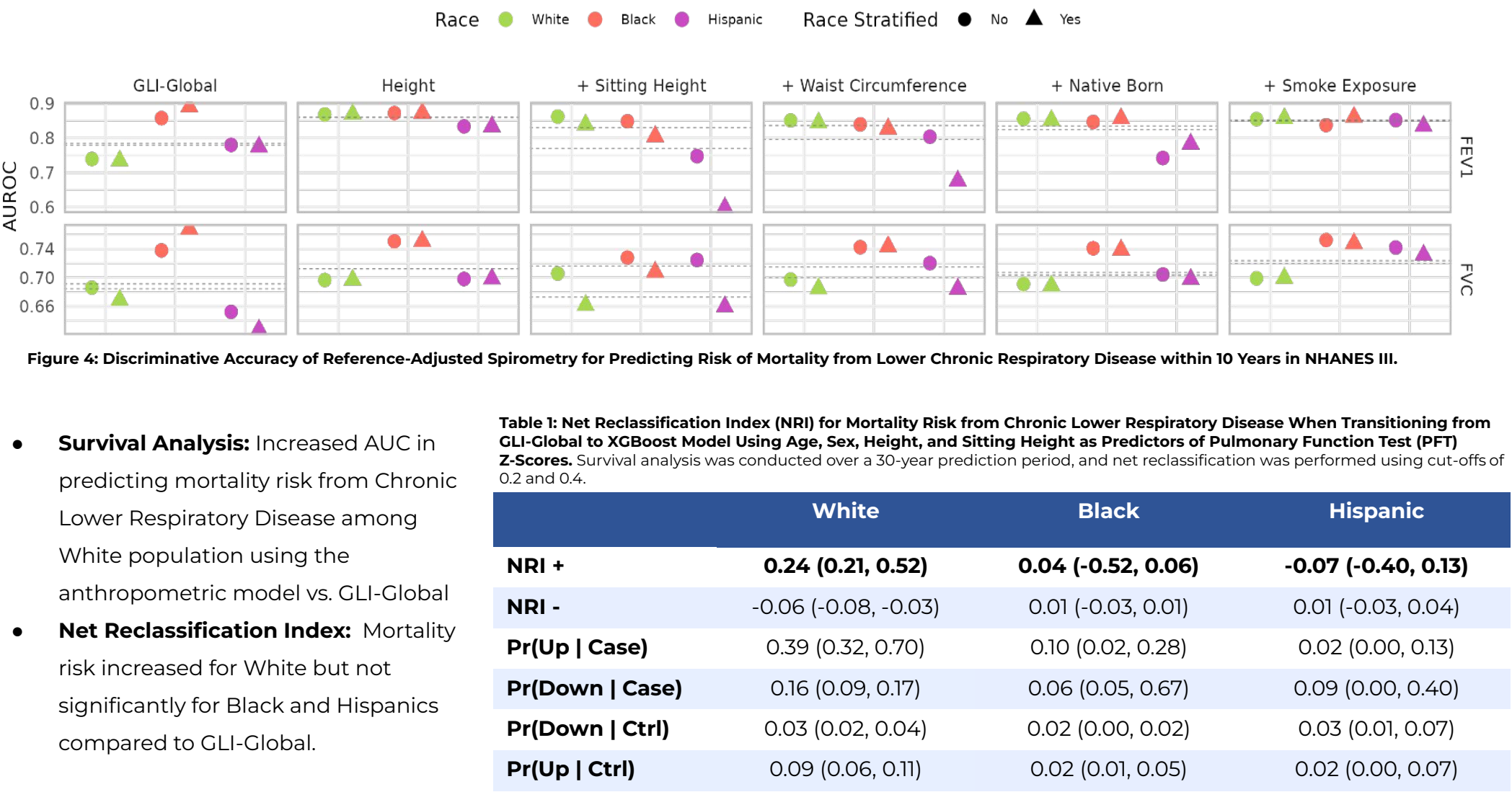
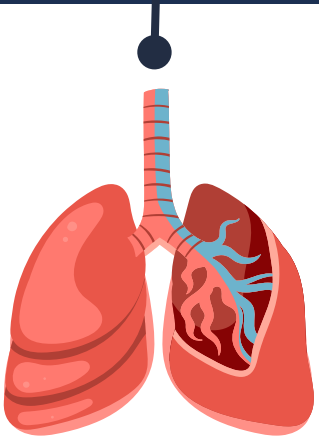


Figure 3: Performance of Race-Free and Race-Based Models for Predicting FVC (mL) on asymptomatic individuals. GLI-Global and five XGBoost models were trained using 80:20 split of UK Biobank. Each model is adjusted by age and sex. Features were added incrementally to each model and were evaluated on UKB test and NHANES. Hispanic was encoded as "Other".

RESPIRATORY OUTCOME PREDICTION



CONCLUSION



Race-Based Difference Explained

Race-based differences in FVC are better explained by anthropometric factors. Exposures failed to explain differences, possibly due to having a larger impact on baseline lung function during development.

Race-Neutral Model Robust to Domain Shift

Race-neutral models outperforms GLI-Global and race-based methods across cohorts. Sitting height and waist circumference measures improve PFT prediction and reduce performance disparities between racial groups.

Improved Prediction of Death from Asthma or COPD

Using anthropometric model, net reclassification for mortality risk increased for White individuals and did not significantly change for Black and Hispanic individuals compared to the GLI-Global model.

Beyond Pulmonary Function Tests

The ARC method expands into different clinical areas where race has been historically influential, potentially providing more precise and fair equations across multiple organ systems.

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Traditionally, PFTs have incorporated factors such as age, sex, height, and race to predict lung function, based on long-standing observations

An international effort was initiated to gather large-scale spirometry data from diverse populations

Data from minority groups was upsampled to ensure they are adequately represented in the model

A bar chart comparing 'Outcomes' for three groups: Majority, Minority Group 1, and Minority Group 2. For each group, two bars are shown: 'Race-Stratified' (dark grey) and 'Race-Neutral with IPW' (blue). The y-axis ranges from 0.00 to 1.00 with increments of 0.25. For the Majority group, the Race-Stratified bar is at approximately 0.90 and the Race-Neutral bar is at approximately 0.70. For Minority Group 1, the Race-Stratified bar is at approximately 0.60 and the Race-Neutral bar is at approximately 0.70. For Minority Group 2, the Race-Stratified bar is at 0.50 and the Race-Neutral bar is at approximately 0.65.

Group	Race-Stratified	Race-Neutral with IPW
Majority	0.90	0.70
Minority Group 1	0.60	0.70
Minority Group 2	0.50	0.65

ARC: ALGORITHMIC APPROACH TO REPLACING RACE CORRECTION

Reproducible Spirometry

Healthy & Asymptomatic

Non-Smoker

Age 40-70

UKBioBank
n=150k

NHANES
n=3k

White Black Asian Hispanic Other

Forward Greedy Search with
Associated Features

$PFT \sim f(\text{Age, Sex, Height, Race, Covariates})$

Race Effect Size

Asian Black Hispanic Other White

Train
UK BioBank

Test
UK BioBank

Test
NHANES

The diagram illustrates the relationship between five categories: Model, Race Stratified, Features, Cohort, and Target. Each category is represented by a vertical bar. The 'Model' bar has two nodes: GAMLSS and XGBoost. The 'Race Stratified' bar has two nodes: True and False. The 'Features' bar has five nodes: Height, Sitting Height, Waist Circumference, Education, and Native Born. The 'Cohort' bar has two nodes: UKBioBank and NHANES. The 'Target' bar has three nodes: FEV1, FVC, and FEV1 / FVC. Blue lines represent connections between nodes across the categories. GAMLSS is connected to True, False, Height, Sitting Height, Waist Circumference, Education, Native Born, UKBioBank, and NHANES. XGBoost is connected to True, False, Height, Sitting Height, Waist Circumference, Education, Native Born, UKBioBank, and NHANES. True is connected to Height, Sitting Height, Waist Circumference, Education, and Native Born. False is connected to Height, Sitting Height, Waist Circumference, Education, and Native Born. Height is connected to UKBioBank and NHANES. Sitting Height is connected to UKBioBank and NHANES. Waist Circumference is connected to UKBioBank and NHANES. Education is connected to UKBioBank and NHANES. Native Born is connected to UKBioBank and NHANES. UKBioBank is connected to FEV1, FVC, and FEV1 / FVC. NHANES is connected to FEV1, FVC, and FEV1 / FVC.

The diagram illustrates the deployment phase of a machine learning pipeline. On the left, a blue human figure is shown interacting with a bar chart and a pie chart, representing the analysis of model performance and clinical/financial implications. On the right, three red-bordered boxes are stacked vertically, containing the text: "Feature Importance", "Performance Metrics", and "Clinical and Financial Implications". A large red arrow points from the bottom of these boxes down to a red box at the bottom labeled "Deployment in Global Cohort".

Deployment in Global Cohort